

CS201 – Data Structures II

Lecture 00

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Outline

- What is this course about?
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- Course outcomes
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What is this course about?

- Sound grounding on basic data structures and their operations
- Understanding of advanced data structures and their amortized analysis
- Understanding of the implementations of the covered data structures
- How to choose a data structure given a problem at hand

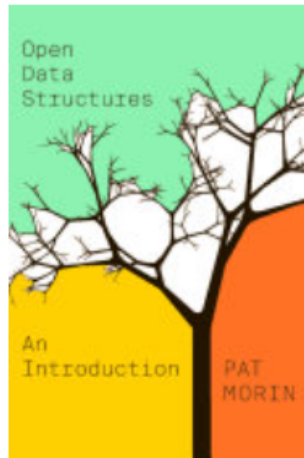
Course Objectives

- We have massive amount of data which is hard to organize
- Data structures help us organize such a huge amount of data and help perform operations on it efficiently
- Understanding of data structures allows for more efficient and more correct code

Course Outcomes

- **Perform** basic operations on elementary data structures: list, tree, stack, queue
- **Design** a system which is robust and scalable perform basic operations on higher level data structures: hash table, self balancing trees, heap, graph, trie, inverted index
- **Implement** any of the covered data structures
- **Implement** a researched data structure
- **Collaborate** on a technical project.

Course Textbooks



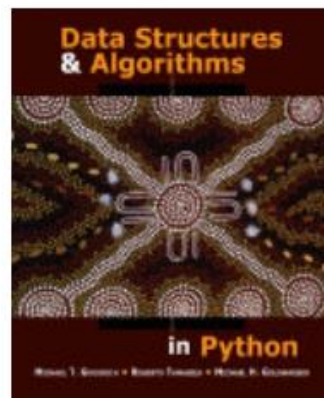
Open Data Structures

ISBN: 9781927356388

Authors: Pat Morin

Publisher: Athabasca University Press

Publication Date: 2013-01-01



Data Structures and Algorithms in Python

ISBN: 9781118476734

Authors: Michael T. Goodrich, Roberto

Tamassia, Michael H. Goldwasser

Publisher: Wiley Global Education

Publication Date: 2013-03-08

Teaching Approach

- Lecture material will be shared before the start of the week
 - You are expected to read the relevant section from the book for the shared material
- In class activities will be carried out to assimilate the concepts

Course Assessments

- Assignments (4)
 - Ensure that all assignments are submitted on or before allotted time
 - Late submission policy applies (see syllabus for details)
- Quizzes (7-8)
 - Spread over the entire duration of course (see syllabus)
- Project (1)
- Final Exam

Assessments Weightage

Instrument	% of grade
Quizzes (7-8)	25%
Assignments (4)	30%
Final Exam (1)	25%
Project (1) Team of 4	20%

Plagiarism Policy

- Plagiarism Policy:
 - **Zero tolerance** towards cheating/plagiarism!!
 - **holds for final exam, assignments, projects and quizzes as well**
- You are expected to submit **your genuine work**.
- If you are using content from elsewhere **a proper reference/citation** should be given
- If you use chatGPT or any other similar AI application, you **must share the prompts** that you used and the content it provided to you clearly **highlighting your contribution**.
- **Failing any of the above, a conduct case may be filed against you so when in doubt please consult your instructor**

Course Syllabus

- Detailed document share on [simple syllabus](#).
- Lets go through it in detail.

Project Details

- To be done in group of four max.
- Details are given on [this page](#).

The screenshot shows a web browser window with the URL <https://hulms.instructure.com/courses/3560/pages/projects>. The browser's address bar and tabs are visible at the top. The page content is for a course titled "CS 201-L1,L2,L3,L4,L5" and is part of the "Spring Semester 2024". The page is titled "Projects" and is marked as "Published". A sidebar on the left contains navigation links: Home, Announcements, Assignments, Discussions, Grades, People, Pages (selected), Files, Syllabus, Outcomes, Rubrics, and Quizzes. The main content area contains the following text:

As part of the course, you are required to develop a project that demonstrates your understanding and implementation of data structures. Your project should showcase your ability to use data structures effectively to solve real-world problems and analyze their performance.

There are several options for your project. You can choose to:

- 1. Implement an advanced data structure:**
Choose a data structure from the provided [list](#) and implement it with some applications. You should be able to demonstrate the usage and functionality of the data structure in your project. You must be able to demonstrate the working, usage, advantages, and disadvantages at the end of the project.
- 2. Compare the performance of different data structures:**
Choose a problem statement and analyze the performance of different data structures to solve it. You should compare the time and space complexities of different data structures and provide insights into their strengths and weaknesses.

Teacher's Contact Details

- Room C-215 Faculty Pod C-203
- Please email me to pick a slot for meeting before approaching me

Teacher's Advice

- Study regularly
- Work hard and be proactive
- Be honest
- Practice

Next Lecture

- Lecture 01